Yingen Zhu

yingen@u.nus.edu | Google Scholar | https://yingen-zhu.github.io

Education

- **National University of Singapore** | Singapore *Aug.2025–Jul.2029 (Expected)*
  PhD in Industrial Design
  Advisor: Prof. Irmandy Wicaksono
- **Southern University of Science and Technology** | Shenzhen, China *Sep.2022–Jun.2025*
  MSc in Electronic Science and Technology
  Advisor: Prof. Wenjin Wang
- **Xiangtan University** | Xiangtan, China *Sep.2018–Jun.2022*
  BSc in Electronic Information Engineering

Selected Publications

Journals (†: Equal contribution):

1. **Y. Zhu**†, Y. Ge†, Q. Wei, et al. "Camera-based Bi-modal PPG-SCG: Sleep Privacy-protected Contactless Vital Signs Monitoring." IEEE Internet of Things Journal. 2024. (JCR Q1, IF=8.2,).
2. **Y. Zhu**, H. Hong, W. Wang. "Privacy-protected Contactless Sleep Parameters Measurement using A Defocused Camera." IEEE Journal of Biomedical and Health Informatics. 2024. (JCR Q1, IF=6.7,).
3. Y. Ge†, **Y. Zhu**†, S. Liaqat, et al. "Implementation of Voxel Selective Ellipse Normalization to Enhance Radar Respiration Estimation in Metallic Chamber." IEEE Internet of Things Journal. 2025. (JCR Q1, IF=8.2,).
4. Y. Zeng, **Y. Zhu**, X. Song, et al. "Camera-based Neonatal Blood Pressure Estimation via Pulse Transit Time - A Proof of Concept in NICU". IEEE Internet of Things Journal. 2025. (JCR Q1, IF=8.2,).
5. D. Huang, Y. Zeng, **Y. Zhu**, et al. "Camera-based Respiratory Imaging System for Monitoring Infant Thoracoabdominal Patterns of Respiration." IEEE Journal of Biomedical and Health Informatics. 2024. (JCR Q1, IF=6.7,).
6. D. Huang, X. Tao, Y. Huang, Y. Wang, **Y. Zhu**, et al. "Camera-based Respiratory Imaging for Intelligent Rehabilitation Assessment of Thoracic Surgery Patients." IEEE Internet of Things Journal. 2024. (JCR Q1, IF=8.2,).
7. Z. Zhou, J. Huang, H. Li, L. Liu, **Y. Zhu**, et al. "Camera Seismocardiogram based Monitoring of Left Ventricular Ejection Time." IEEE Transactions on Biomedical Engineering. 2025. (JCR Q2, IF=4.4,).

Conferences:

1. **Y. Zhu**, D. Huang, Y. Zeng, et al. "Occlusion-robust Sleep Posture Detection using Body Rolling Motion in a Video." The 45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society. 2023. ().
2. **Y. Zhu**, J. Huang, H. Lu, et al. "Privacy Protected Contactless Cardio-respiratory Monitoring using Defocused Cameras during Sleep." IEEE International Conference on E-health Networking, Application & Services. 2023. ().
3. S. Yi, D. Yu, **Y. Zhu**, et al. "Illumination-robust Camera-PPG via Skin-Guided Auto-Exposure." The 46th Annual International Conference of the IEEE Engineering in Medicine & Biology Society. 2024. ().

Patents:

1. W. Wang, **Y. Zhu**, H. Mo, et al. "Magnetic resonance image rearrangement method, system, and storage medium." CN granted patent: CN119355609B. 2025.
2. W. Wang, **Y. Zhu**, H. Mo, et al. "Camera-seismocardiography based magnetic resonance gating method, device, terminal, and medium." CN granted patent: CN119355608A. 2025.
3. W. Wang, **Y. Zhu**, Q. Wei. " Multi-camera based PTT estimation method, system, terminal, and medium." CN granted patent: CN119184647A. 2024.
4. W. Wang, **Y. Zhu**. "A multi-wavelength pulse transit time estimation method, system, terminal, and medium." CN granted patent: CN119174596B. 2024.
5. Y. Chen, Q. Huang, X. Chen, H. Lin, J. Peng, **Y. Zhu**, Y. Li. "A system and method for identifying dangerous objects on balconies using OpenMV-based Unmanned Aerial Vehicle." CN granted patent: CN111947603B. 2022.

Honors & Awards

- Top 10 Graduates with a Postgraduate Degree *2025*
- National 2nd Prize, Healthcare Artificial Intelligence Competition *2024*
- National Scholarship for Graduate Students *2024*
- National 2nd Prize, The National "Challenge Cup" Technological Innovation Competition *2022*
- Great Leader Memorial Scholarship | Top 1%, 1/220 *2021*
- 2nd Prize in Hunan Province, The Smart Car and Robotics Competition *2021*
- 3nd Prize in Hunan Province, The National "Beidou Cup" Technological Innovation Competition *2021*
- 2nd Prize in the Central-Southern Region, The National "Beidou Cup" Technological Innovation Competition *2020*

Teaching Experience

- TA, Fiberpunk *Spring 2026*
- TA, Soft Devices *Fall 2025*

Academic Service

- Conference reviewer: ICHEC
- Journal reviewer: IEEE JBHI, IEEE IoT-J

Skills

- Programming: MATLAB, Python, C/C++, Swift
- Languages: Mandarin (native) and English (fluent)
- Others: PCB Assembly & Design, Rihno, Unity, Blender